Department of Mechanical and Aerospace Engineering

ME5103 Finite Element Methods

Lab 4: CST Element and FEM Assembly

16-10-2026

Report Template: Use similar format as earlier reports.

Submission Requirements

- MATLAB code
- Nodal displacement results
- Tip displacement comparison
- Brief discussion

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1. Single CST Element

Consider a triangular element with nodes:

$$(0, 0), (1, 0), (0, 1)$$

Material properties

$E = 200 \times 10^3$ MPa, $\nu = 0.3$, $t = 1$

Boundary Conditions

- Node 1: $u = v = 0$
- Node 2: $v = 0$

Loading

$F_y = -100$ N at node 3

Tasks

- Compute area, B matrix and D matrix
- Form element stiffness matrix
- Solve for nodal displacements
- Compute strain and stress

2. Patch Test

Square domain:

$$(0, 0), (1, 0), (1, 1), (0, 1)$$

Elements

- Element 1: (1–2–3)
- Element 2: (1–3–4)

Prescribed displacement field

$$u = 0.001x, \quad v = 0.002y$$

Tasks

- Compute nodal displacements

- Compute strain in each element
- Compare with exact strain

3. Cantilever Beam using CST Elements

A rectangular cantilever beam is to be modeled using CST elements.

Geometry

•Length $L = 1$ m

•Height $H = 0.1$ m

Material Properties

$E = 210 \times 10^3$ MPa, $v = 0.3$, $t = 1$

Boundary Conditions

Left edge is fixed: $u = 0$, $v = 0$

Loading

$F_y = -100$ N applied at the tip node

Mesh Requirement

•Divide beam length into at least 10 segments

•Divide height into 1–2 segments

•Generate node coordinates programmatically

•Divide each rectangular cell into two triangular elements

•Optional: provision to increase number of segments in the x and y direction to study convergence

Tasks

•Generate structured mesh (nodes and connectivity)

•Assemble global stiffness matrix

•Apply boundary conditions

•Solve for nodal displacements

•Determine vertical displacement at beam tip

•Perform mesh refinement study

Analytical Solution

$$\delta_{tip} = \frac{FL^3}{3EI}, \quad I = \frac{H^3}{12}$$

Discussion

•Compare FEM and analytical results

•Comment on convergence with mesh refinement

•Explain limitations of CST in bending-dominated problems